
SUMMARY — I am a Senior Research Scientist at Beink Dream, where I lead AI initiatives from fundamental research to production deployment—designing and scaling AI systems for 2D and 3D applications. Previously, I worked at CEA (France) and earned my Ph.D. from École Normale Supérieure under the supervision of Prof. Nelly Pustelnik. My work focuses on bridging physics and AI, with a particular emphasis on efficient model design. I am especially driven by the challenge of learning compact, physically grounded representations from continuous high-dimensional signals, with strong connections to modern physics research.

EXPERIENCES

Beink Dream

Mar 2025 – Present

Senior Research Scientist

Paris, France

Architected end-to-end agentic AI systems integrating LLM reasoning, generative vision models, and structured 3D/CAD output — building multi-modal systems that maintain a structured internal representation of scene state across 2D, 3D, and CAD modalities.

CEA – DIGITEO Labs

Feb 2024 – Feb 2025

Research Engineer

Saclay, France

Worked on lightweight learning models for 3D CT reconstruction — physics-consistent scene representations from sparse, noisy measurements and limited high-dimensional observations.

INMA, ICTEAM – UC Louvain

Sep 2020 – Aug 2021

Visiting PhD Researcher

Louvain, Belgium

Designed scalable accelerated algorithms to solve joint image restoration and edge detection on large-scale physics experiment images.

OGoxe – Météo France

Jan 2020 – Aug 2020

Research Intern

Toulouse, France

Built velocity estimation models from video sequences — learning to infer physical dynamics (fluid flow) from continuous visual observations, an early instance of physics-grounded world dynamics modeling.

IMFT – Toulouse Fluid Mechanics Institute

Jul 2019 – Oct 2019

Research Intern – High Performance Computing

Toulouse, France

Worked on a novel Bucketing-based algorithm for particle-based simulation using parallelized mesh-less models across multi-GPU clusters.

EDUCATION

École Normale Supérieure de Lyon

Sep 2020 – Dec 2023

Ph.D. in Applied Mathematics, AI, Signal and Image Processing

Lyon, France

Thesis: *Variations on the Mumford-Shah functional for interface detection in degraded images: from proximal algorithms to unrolled architectures.*

Advised by Prof. Nelly Pustelnik & Dr. Marion Foare.

INSA Toulouse

Sep 2015 – Sep 2020

Master of Engineering in Applied Mathematics

Toulouse, France

Major in Mathematical Modelling and Numerical Analysis.

PUBLICATIONS

Unfolded Discrete Mumford-Shah Functional for Joint Image Denoising and Edge Detection, *EUSIPCO 2025*

H.T.V Le, Marion Foare, Audrey Repetti, Nelly Pustelnik

Proximal Neural Networks based Reconstruction for Few-View CT Applications, *ICT 2025*

H.T.V Le, C. Bossuyt, J. Escoda, H. Wang, J. De Beenhouwer, J. Sijbers

Embedding Blake-Zisserman Regularization in Unfolded PNNs for Enhanced Edge Detection, *Under review*

H.T.V Le, Marion Foare, Audrey Repetti, Nelly Pustelnik

Unfolded Proximal Neural Networks for Robust Image Gaussian Denoising, *IEEE TIP 2024*

H.T.V Le, Audrey Repetti, Nelly Pustelnik

Proximal Based Strategies for Solving Discrete Mumford-Shah With Ambrosio-Tortorelli Penalization on Edges, *IEEE SPL 2022*

H.T.V Le, Marion Foare, Nelly Pustelnik

The Faster Proximal Algorithm, the Better Unfolded Deep Learning Architecture? A Study Case of Image Denoising, *EUSIPCO 2022*

H.T.V Le, Nelly Pustelnik, Marion Foare

Fast Unfolded Proximal Algorithms for the Analysis of Piecewise Homogeneous Fractal Images, *Gretsi 2022*

H.T.V Le, Barbara Pascal, Nelly Pustelnik, Marion Foare, Patrice Abry

ACTIVITIES

Teaching: Signal processing, convex optimization, unfolded NNs and inverse problems (ENS Lyon); Signal processing and Cryptography Mathematics (EPITA Lyon); Fundamental Machine Learning (EPITA Paris)

Talks:

EUSIPCO, Belgrade, Serbia

Aug 2022

Conference on Digital Geometry and Discrete Variational Calculus

Mar 2021

Open-Source Software:

P-DMS-AT [Python] - Proximal strategies for Discrete Mumford-Shah with Ambrosio-Tortorelli edge penalization.

Proximal Neural Networks [PyTorch] — Lightweight unrolled architectures combining iterative algorithms and deep networks for inverse problems.

BZ-PNN [PyTorch] — Super-lightweight model for enhanced edge detection via Blake-Zisserman regularization.

DMS-PNN [PyTorch] — Unfolded Discrete-Mumford-Shah Proximal Neural Networks.

Bucketing-based Algorithm [Fortran, C/C++, Python] — Parallelized linearly transformed particle method for diffusion equations.

AWARDS

- Third prize, Vietnam National Physics Olympiad **2015**
- Fourth prize, Vietnam National Physics Olympiad **2014**
- Silver medal, Northern Regional Vietnam Physics Olympiad **2013**